

Coleco followed up on its previous offering by releasing the ColecoVision in 1982, a console that promised a more powerful experience closer to many of the popular arcade games systems. The initial offerings of games bundled with the ColecoVision further helped boost its popularity, with titles including Nintendo's Donkey Kong as the pack-in cartridge, Zaxxon, and a few other less popular titles, such as Lady Bug, Cosmic Avenger, and Venture. Many titles released later, like titles from the Star Trek franchise, also Tarzan and Popeye, to name a few.

In response to the ColecoVision, Atari released its 5200 SuperSystem, which drew inspiration from the graphics and audio chips found in Atari's 400 personal computer. However, this system provided very incremental improvements to the 2600, and games released for this version were just modified versions of the 2600 games; this lack of new titles and storylines led to consumers losing interest in Atari's offerings.

In addition to bulky gaming consoles that could be used in multiplayer mode, the second generation was also witness to the rising adoption of hand-held consoles. While those released in this period were still in their infancy, they helped lay the groundwork for some iconic consoles released later.

The first programmable game system at this time was the MicroVision, created by Milton Bradley. The console allowed interchangeable cartridges, and combined portability and ease of use as its defining features. Unfortunately, despite roaring initial sales, the lack of gaming cartridges, its small and unreliable screen, and lack of support for existing game developers led to the downfall of this pioneering device.

In Japan, Epoch Systems released its Game Pocket Computer in 1984. However, despite a sizeable LCD screen and resulting graphics comparable to the Atari 2600, the system was a failure in Japan and wasn't eventually released in North America.

Before the Game Pocket Computer, however, debuted Nintendo's Game & Watch series of handheld gaming consoles. Game designer Gunpei Yokoi created each Game & Watch to feature a single game that was played on a segmented LCD screen that was pre-printed with an overlay, in addition to a clock, an alarm, or both. The pre-printing was what allowed the Game & Watch to allow only one game to be played at a time. However, this didn't stop the rising popularity of the device, and its versatile titles such as Donkey Kong, The Legend of Zelda and Mario Bros only added to its appeal. Another note-worthy feature of the device was its usage of the modern "cross" D-pad design, which was later patented and received multiple awards.